

THA TUM, Thailand

WMO: 484160

Lat: 15.316N

Lon: 103.680E

Elev: 129

StdP: 99.78

Time Zone: 7.00 (E07)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.2	16.6	10.4	8.0	18.2	11.8	8.7	19.2	5.9	23.4	4.8	22.7	0.7	50	0.319

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	9.9	38.0	25.9	36.9	25.9	35.9	25.9	28.0	32.6	27.6	32.3	27.3	32.1	1.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.9	22.9	30.1	26.5	22.4	29.9	26.2	21.9	29.7	90.4	32.5	89.0	32.4	87.7	32.4	30.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.0	4.1	3.4	DB	13.0	39.4	1.2	1.2	12.2	40.2	11.4	40.9	10.7	41.6	9.9	42.5	(4)
			WB	10.9	28.9	1.5	0.6	9.8	29.3	9.0	29.7	8.1	30.0	7.1	30.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.0	24.5	26.7	29.3	30.7	30.2	29.8	29.1	28.8	28.3	27.8	26.5	24.4
	DBStd	2.80	2.51	2.88	2.59	2.06	1.56	1.24	1.27	1.08	1.13	1.14	2.05	2.57
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	1	0	0	0	0	0	0	0	0	0	0	0	1
	CDD10.0	6574	449	468	597	622	626	596	592	583	549	552	494	447
	CDD18.3	3533	191	234	339	372	368	346	334	324	299	293	244	189
	CDH23.3	40350	1698	2539	4227	4937	4682	4311	3946	3746	3273	3082	2297	1612
	CDH26.7	18107	666	1219	2259	2740	2357	2022	1662	1484	1168	1097	862	571
Wind	WSAvg	0.8	0.9	0.7	0.7	0.6	0.7	0.9	0.8	0.8	0.5	0.7	1.1	1.1
Precipitation	PrecAvg	1385	9	15	47	83	172	195	224	235	263	119	22	2
	PrecMax	1774	81	93	182	206	297	316	381	384	514	275	111	19
	PrecMin	1127	0	0	1	35	86	81	46	61	119	1	0	0
	PrecStd	162	20	21	43	46	60	65	87	78	90	69	28	4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.4	37.1	38.9	39.8	38.3	36.7	35.9	34.6	34.0	33.4	34.7	33.7
		MCWB	23.7	25.0	25.1	25.9	26.8	26.6	26.5	26.9	26.7	26.1	25.7	24.7
	2%	DB	33.0	35.6	37.5	38.4	36.8	35.4	34.3	33.5	32.9	32.4	33.2	32.3
		MCWB	23.6	24.8	25.3	26.1	26.7	26.7	26.7	26.7	26.7	25.9	25.4	24.2
	5%	DB	31.8	34.4	36.4	37.2	35.6	34.4	33.4	32.7	32.2	31.8	32.1	31.2
		MCWB	23.6	24.6	25.3	26.0	26.6	26.6	26.7	26.6	26.5	25.7	25.1	23.7
	10%	DB	30.5	33.0	35.1	35.9	34.4	33.5	32.5	32.0	31.4	31.1	31.1	29.9
		MCWB	23.0	24.3	25.3	26.0	26.5	26.6	26.5	26.5	26.4	25.6	24.5	23.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.9	27.3	27.6	28.0	28.1	28.1	28.2	28.2	28.1	27.7	27.3	26.6
		MCDB	30.7	33.5	33.2	34.9	33.8	32.9	32.3	31.8	31.6	30.9	31.5	30.4
	2%	WB	25.4	26.4	26.9	27.4	27.6	27.6	27.6	27.7	27.5	27.2	26.6	25.7
		MCDB	30.4	32.9	33.8	34.3	33.4	32.6	32.0	31.4	31.1	30.6	31.0	30.1
	5%	WB	24.5	25.6	26.3	26.9	27.2	27.2	27.2	27.2	27.1	26.7	26.1	24.8
		MCDB	30.0	32.1	33.6	33.9	33.0	32.3	31.7	31.1	30.8	30.1	30.4	29.7
	10%	WB	23.6	24.9	25.8	26.5	27.0	27.0	26.8	26.9	26.8	26.4	25.5	24.0
		MCDB	29.0	31.1	33.1	33.4	32.7	32.1	31.3	30.8	30.2	29.7	29.5	28.6
Mean Daily Temperature Range	5% DB	MDBR	11.8	12.0	11.0	9.9	8.3	7.4	6.8	6.5	6.2	6.8	8.7	10.4
		MCDBR	12.4	12.7	12.1	11.1	9.8	8.6	8.0	7.5	7.2	7.5	9.6	11.1
		MCWBR	4.9	4.3	3.4	2.9	2.7	2.5	2.7	2.7	2.7	3.0	3.6	4.5
	5% WB	MCDBR	11.1	11.1	10.7	10.0	8.6	7.8	7.3	6.9	6.7	6.8	8.4	10.2
	MCWBR	5.6	4.7	4.0	3.3	3.0	2.8	3.0	2.9	2.9	3.1	3.9	5.3	
Clear-Sky Solar Irradiance	taub	0.488	0.608	0.668	0.612	0.508	0.469	0.462	0.464	0.503	0.524	0.475	0.466	
	taud	2.057	1.763	1.674	1.839	2.162	2.282	2.302	2.287	2.167	2.079	2.192	2.156	
	Ebn at Noon	800	722	693	734	802	825	833	840	810	781	808	810	
	Edh at Noon	162	226	253	215	153	134	132	136	153	163	140	143	
All-Sky Solar Radiation	RadAvg	5.02	5.26	5.34	5.72	5.68	5.34	5.07	4.88	4.70	5.03	4.98	4.85	
	RadStd	0.27	0.23	0.20	0.23	0.30	0.22	0.34	0.32	0.34	0.34	0.20	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.44	N/A	N/A	N/A	+0.32	+0.38	N/A	N/A	N/A	N/A
Regional (17 neighbors)		+0.27	N/A	N/A	N/A	+0.23	+0.21	N/A	N/A	+105	+105

Nomenclature: See separate page